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Component 1: Image Processing

* Input: coronagraph observations

□ Download sequential polarized images

□ Apply calibration

□ Perform background subtraction

□ Combine sequential images to generate pB images

□ Stacking & median filtering

* Output: representative pB image

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Component 2: Synthetic Image Generation

* Input: coronagraph observation headers

□ Generate MHD model solution

□ Coordinate transformation

□ Generate synthetic pB image using FORWARD

□ Coalignment using .fits headers

* Outputs: synthetic images (coaligned) and MHD model outputs (B, n_e parameters)

output representative pB image

output synthetic pB image

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Component 3: Image Segmentation

* Inputs: representative image and synthetic image

□ Preprocessing & noise reduction

□ Transform to plane polar coordinates

□ Detrending & thresholding

□ Measure detected features as polynomials

* Outputs: detected features & geometry parameters

output feature geometry parameters

output MHD
model parameters
(B, n_e parameters)

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Component 4: Comparative Analysis

* Inputs: MHD model B field and detected features

■ Compare orientation angles (real vs synthetic) vs model B field orientation

□ Calculate angle differences

□ Perform statistical analysis on angle differences

□ Generate metrics and global statistics

* Outputs: comprehensive metrics and statistics